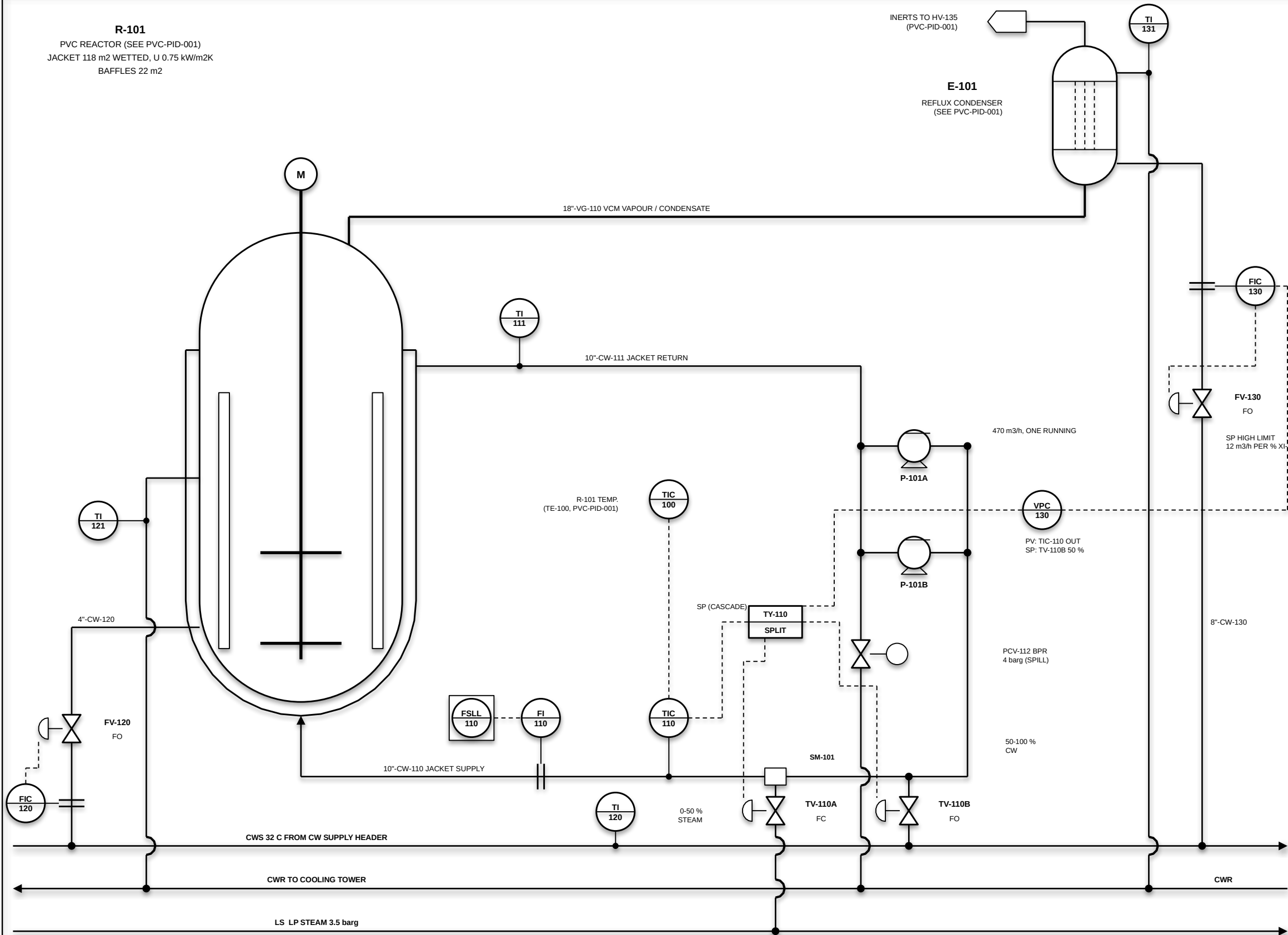




GENERAL NOTES:

- SEE PVC-PID-001 NOTES 1-3 FOR THE DESIGN BASIS AND RECIPE, NOTE 8 AND THE SIS TABLE FOR TRIPS. THIS SHEET SHOWS THE R-101 HEAT-REMOVAL SYSTEMS AND THE TEMPERATURE CONTROL.
- JACKET: CLOSED TEMPERED-WATER LOOP, P-101A/B 470 m3/h (ONE RUNNING; B STARTED BY OPERATOR - AUTO-START TBC). CW 32 C INJECTED THROUGH TV-110B (FAIL OPEN) AND LP STEAM THROUGH TV-110A (FAIL CLOSED) + STEAM MIXER SM-101; THE SAME MASS SPILLS TO CWR THROUGH BACK-PRESSURE REGULATOR PCV-112 (LOOP 4 barg). FSL-110 < 120 m3/h FOR 30 s WHILE REACTING -> SIS I-4 (SHORT-STOP).
- TEMPERATURE CONTROL: TIC-100 (MASTER, R-101 CONTENTS; Kc 10, Ti 1500 s, Td 120 s) SETS TIC-110 (SLAVE, JACKET INLET; Kc 2, Ti 40 s). TY-110 SPLIT RANGE: OUTPUT 0 % = FULL STEAM, 50 % = BOTH SHUT, 100 % = FULL CW.
- VPC-130 = VALVE-POSITION CONTROL: HOLDS TV-110B NEAR 50 % OPEN BY RESETTING FIC-130, SO THE JACKET KEEPS ITS TRIM MARGIN AND THE REFLUX CONDENSER TAKES THE GROWING LOAD. FIC-130 SP HIGH LIMIT = 12 m3/h PER % OF XI-100 (ANTI-FOAM: TOO MUCH BOIL-UP BEFORE ~10 % CONVERSION CARRIES FOAM AND PVC INTO E-101).
- BAFFLES: FOUR ADJUSTABLE WATER-COOLED BAFFLES (22 m2) ON ONCE-THROUGH CW, FIC-120 80 m3/h.
- DESIGN DUTIES AT 2 h (X = 26 %): REACTION 2.71 MW + AGITATOR 0.29 MW = JACKET 1.48 + BAFFLES 0.40 + E-101 0.98 MW. PEAK REACTION 3.43 MW NEAR X = 0.7; THE JACKET ALONE (MAX ~1.8 MW) CANNOT CARRY IT - E-101 ADDS THE MISSING AREA.
- CW SUPPLY 32 C (TROPICAL DESIGN). E-101 CW RETURN APPROACHES 55 C AT LOW FLOW: MINIMUM-FLOW / SCALING LIMIT TBC WITH THE CW TREATMENT VENDOR.
- LINE SIZES AND MATERIALS PRELIMINARY - TBC BY PIPING. SERVICE CODES AS PVC-PID-001; LS = LP STEAM.

LEGEND			
	PROCESS LINE		UTILITY LINE (CW, STEAM, N2)
	SIGNAL LINE		LINE JUMP (NO JOIN)
	CONTROL VALVE FO / FC		ON/OFF VALVE (SEQ. / SIS)
	MANUAL VALVE		ORIFICE FLOW ELEMENT
	DCS INSTRUMENT		FIELD INSTRUMENT
	SIS FUNCTION		DCS FUNCTION BLOCK
	OFF-SHEET LINK		RELIEF VALVE

CONTROL LOOP SCHEDULE				
LOOP	MEASURES	MOVES	FAIL	SET / NOTE
FIC-101	WATER CHARGE	FV-101	FC	240 m3/h, FQ 60 t
FIC-102	VCM CHARGE	FV-102	FC	180 t/h, FQ 51 t
LIC-100	R-101 LEVEL	FIC-103 SP	-	83 %
FIC-103	INJECTION WATER	FV-103	FC	CAS (~3 m3/h)
TIC-100	R-101 TEMP.	TIC-110 SP	-	57.0 C (K 67)
TIC-110	JACKET INLET	TY-110	-	CAS (~38 C)
TY-110	0-50 % OUT	TV-110A STEAM	FC	HEAT-UP
TY-110	50-100 % OUT	TV-110B CW	FO	COOLING
FIC-120	BAFFLE CW	FV-120	FO	80 m3/h
VPC-130	TIC-110 OUT	FIC-130 SP	-	50 % (NOTE 5)
FIC-130	E-101 CW	FV-130	FO	CAS
HIC-135	-	HV-135 INERTS	FC	MANUAL
HIC-105	-	PV-105 DEGAS	FC	END OF BATCH

SIS INTERLOCKS			
I-	INITIATOR	SETTING	ACTION
1	TSHH-101 R-101 TEMP.	> 63 C, 10 s	A
2	PSHH-101 R-101 PRESS.	> 12.5 barg, 5 s	A
3	XS-101 AGITATOR STOP	REACTING, 15 s	A
4	FSL-110 JACKET FLOW	< 120 m3/h, 30 s	A
5	LSHH-100 R-101 LEVEL	> 94 %, 5 s	B
6	HS-001 ESD PUSHBUTTON	MANUAL	A + XV-108

A = SHORT-STOP XV-104, CLOSE XV-101/102/103, FV-103 TO 0, FORCE TV-110B / FV-120 / FV-130 OPEN. B = CLOSE XV-101/102, FV-103.

ITEM (OTS DESIGN POINT)	VALUE
WATER / VCM CHARGE	60 t (55 C) / 51 t (25 C)
REACTION	57.0 C, 8.6 barg
LEVEL / CONTENTS AT 2 h	83 % / 117.8 t
Q REACTION AT 2 h	2.71 MW (PEAK 3.43)
JACKET / BAFFLES / E-101	1.48 / 0.40 / 0.98 MW
END POINT	dP 1.8 bar, X 0.85
PVC PER BATCH / TIME	43.2 t / 5.3 h

REV	DESCRIPTION	DATE	
A	ISSUED FOR OTS DEVELOPMENT (REVIEW)	25-SEP-26	
CHULA CHEM. ENG. - PVC REACTOR VER 1 (OTS)			
P&ID - R-101 HEAT REMOVAL: JACKET, BAFFLES, E-101 CW			
DRAWN: APINAN SOOTTITANTAWAT	CHK'D: --	APP'D: --	DATE: 25-SEP-26
DWG NO. PVC-PID-002	REV A	SCALE N.T.S.	SHEET 2 OF 2